

230X - RIT Algorithmic Trading

Matthieu Pascal

a. What was your strategy going in?

My strategy was a conservative form of market making on CRZY_M and CRZY_A. Rather than predicting the direction of the stock, I aimed to buy below an estimated fair value and sell above it using LIMIT orders. The algorithm used the order books and recent transactions to estimate a reference price. It treated large displayed quantities cautiously, since those orders could disappear before execution.

The basic quoting principle was:

$$P_{\text{buy}} = \hat{V} - \delta, \quad P_{\text{sell}} = \hat{V} + \delta,$$

where $\hat{V}$ is the estimated fair value and δ is a margin intended to compensate for uncertainty and adverse selection. If both sides executed for the same quantity q , the resulting profit, with no trading fees, would be:

$$\Pi = q(P_{\text{sell}} - P_{\text{buy}}) = 2q\delta.$$

In practice, the algorithm used a valuation interval $[V_{\text{low}}, V_{\text{high}}]$ rather than a single estimate. It placed buys below the lower bound and sells above the upper bound, adjusting the quotes to the available bids and asks. It reduced order size or avoided new orders when conditions appeared unstable.

The objective was to capture short-term fluctuations while limiting inventory exposure. This was not risk-free arbitrage: one side could execute without the other, leaving a position exposed to market movements.

b. What happened during the simulation? What was your size strategy?

I used very small orders, with an entry cap of 25 shares. The algorithm could reduce this quantity further depending on available liquidity, recent trading volume, uncertainty, and remaining inventory capacity. The idea was to limit exposure rather than scale aggressively based on large displayed orders.

The strategy did not work as expected, so I stopped it relatively early. Its conservative sizing limited the financial impact, but I did not run it long enough to establish whether it could be consistently profitable.

After stopping, a residual position remained open. The algorithm required a flat account with no outstanding orders before starting, so this position prevented me from restarting it. This exposed a weakness in the shutdown and recovery process: using LIMIT orders for exits did not guarantee that the account would be flat.

Python script: `rit_adaptive.py`, submitted separately.

c. MARKET versus LIMIT orders and their risks

A MARKET order prioritizes execution over price. It can execute at a worse price than the one observed when the trading decision was made, especially if liquidity is thin or prices move quickly. For a two-sided strategy, slippage between the two executions can eliminate the expected spread or turn it into a loss.

A LIMIT order specifies the highest acceptable purchase price or the lowest acceptable sale price. It controls the execution price but does not guarantee execution. Orders can remain unfilled or execute only partially. If only one side fills, the strategy is left with directional exposure.

LIMIT orders also face adverse selection: a buy order may fill just as the market falls, or a sell order just as it rises. Therefore, obtaining the requested price does not necessarily mean obtaining a favorable trade.

My algorithm used LIMIT orders for both entries and exits. This provided price control, but it also created the risk of leaving a residual position when stopping, which occurred during my session.

d. What was your P&L?

My final P&L was **-\$3.42**. The loss was small because my positions were very small and I stopped early. However, the strategy did not perform as intended, and the residual position prevented me from restarting it. My main takeaway was that reliable execution and position management were just as important as the quoting rule.